

Third Section.

Transformation of the Theta-Functions.

§ 27. The transformation principle.

We return to the defining equations, given in § 17, for the T -functions of m -th order, and, for a reason which will presently be evident, provide the letters ω, a, b, m occurring there with primes, so that these equations read

$$\begin{aligned} T(u + \omega'_1) &= e^{-\pi i[a'_1(2u + \omega'_1) + b'_1]} T(u), \\ T(u + \omega'_2) &= e^{-\pi i[a'_2(2u + \omega'_2) + b'_2]} T(u), \end{aligned} \quad (1)$$

$$a'_2 \omega'_1 - a'_1 \omega'_2 = m'. \quad (2)$$

If a, c are arbitrary integral (positive or negative) numbers, then, by replacing n_1, n_2 in § 17, (16), by $-c, a$, one obtains

$$\begin{aligned} T(u - c\omega'_1 + a\omega'_2) \\ = e^{-\pi i(-ca'_1 + aa'_2)(2u - c\omega'_1 + a\omega'_2) - \pi i(-cb'_1 + ab'_2 - m'ac)} T(u), \end{aligned} \quad (3)$$

an equation which also follows from one of equations (1), if in it a', b', ω' are replaced by

$$-ca'_1 + aa'_2, \quad -cb'_1 + ab'_2 - m'ac, \quad -c\omega'_1 + a\omega'_2.$$

Herein is contained the principle of transformation of the T -functions.

Let b, ∂ be two other integral numbers for which the determinant

$$n = a\partial - bc \quad (4)$$

has a positive value. We put

$$\omega_1 = \partial\omega'_1 - b\omega'_2, \quad \omega_2 = -c\omega'_1 + a\omega'_2, \quad (5)$$

and consequently

$$n\omega'_1 = a\omega_1 + b\omega_2, \quad n\omega'_2 = c\omega_1 + \partial\omega_2. \quad (6)$$

It then has, as one sees from

$$\frac{\omega'_2}{\omega'_1} = \frac{c\omega_1 + \partial\omega_2}{a\omega_1 + b\omega_2}$$

by separating the real and imaginary parts, the imaginary part of $\omega'_2 : \omega'_1$ the same sign as that of $\omega_2 : \omega_1$, namely the positive sign.

We set

$$a_1 = \partial a'_1 - b a'_2, \quad a_2 = -c a'_1 + a a'_2, \quad (7)$$

$$b_1 = \partial b'_1 - b b'_2 - m' b \partial, \quad b_2 = -c b'_1 + a b'_2 - m' a c. \quad (8)$$

From (3) one then concludes that the function $T(u)$ satisfies not only conditions (1), but also the equations obtained from (1) by interchanging $\omega'_1, \omega'_2, a'_1, a'_2, b'_1, b'_2$ with $\omega_1, \omega_2, a_1, a_2, b_1, b_2$, that is, the equations (1) of § 17. Thus it is simultaneously a T -function of the periods ω'_1, ω'_2 and of the periods ω_1, ω_2 , which we indicate by the equation

$$T'(u, \omega'_1, \omega'_2) = T(u, \omega_1, \omega_2). \quad (9)$$

But from (5) and (7) one has

$$a_2\omega_1 - a_1\omega_2 = (a'_2\omega'_1 - a'_1\omega'_2)(a\partial - bc),$$

and hence, if m is the order of T ,

$$m = m'n. \quad (10)$$

According to (8), the characteristic (g_1, g_2) of T , if (g'_1, g'_2) is that of T' (§ 17), is

$$(g_1, g_2) = (\partial g'_1 - bg'_2 - m'b\partial, -cg'_1 + ag'_2 - m'ac). \quad (11)$$

By the transformation of the T -functions one understands the representation of the functions T' with the periods ω'_1, ω'_2 by T -functions with the periods ω_1, ω_2 .

The numbers a, b, c, ∂ are called the transformation numbers, and $n = a\partial - bc$ the degree of transformation.

In order to survey the form of this representation more clearly, we seek the conditions under which $T'(u, \omega_1, \omega_2)$ becomes a Θ -function of the m -th order $\Theta(u, \omega)$ (§ 20). We take b'_1, b'_2 , and consequently also b_1, b_2 , as integers, so that b_1, b_2, b'_1, b'_2 may be replaced by g_1, g_2, g'_1, g'_2 . Then one must put

$$\omega_1 = 1, \quad \omega_2 = \omega, \quad a_1 = 0, \quad a_2 = m,$$

and therefore, by (6), (7), (10),

$$\omega'_1 = \frac{a + b\omega}{n}, \quad \omega'_2 = \frac{c + \partial\omega}{n}, \quad a'_1 = m'b, \quad a'_2 = m'\partial.$$

The function $T'(u, \omega'_1, \omega'_2)$ therefore satisfies the conditions (1):

$$T'(u + \omega'_1) = (-1)^{g'_1} e^{-\pi i m' b (2u + \omega'_1)} T'(u),$$

$$T'(u + \omega'_2) = (-1)^{g'_2} e^{-\pi i m' \partial (2u + \omega'_2)} T'(u),$$

and from this it follows that the product

$$e^{\pi i m' b u^2 / \omega'_1} T'(u, \omega'_1, \omega'_2)$$

is a Θ -function of order m' , with the arguments

$$\frac{u}{\omega'_1}, \quad \frac{\omega'_2}{\omega'_1}$$

and the characteristic (g'_1, g'_2) . We can express this in the equation

$$e^{-\frac{\pi i m' n b u^2}{a + b\omega}} \Theta_{g'_1, g'_2}^{(m')} \left(\frac{nu}{a + b\omega}, \frac{c + \partial\omega}{a + b\omega} \right) = \Theta_{g_1, g_2}^{(m'n)}(u, \omega), \quad (12)$$

where the characteristics are determined by (11). The means for representing these functions are contained in § 21.

We designate the transformation of T and T' by a single letter S , or by (T', T) , thus

$$S = (T', T).$$

If S' denotes a second transformation, by which T' passes into T'' , thus

$$S' = (T'', T'),$$

then from this we can derive a new transformation S'' , by which T passes into T'' . This is called the transformation compounded from S and S' and is denoted by

$$S'' = S'S$$

or

$$(T'', T) = (T'', T')(T', T).$$

With this composition the commutative law does not hold in general; thus SS' is different from $S'S$. But the associative law holds, and is expressed by the formula

$$(T''', T'')[(T'', T')(T', T)] = [(T''', T'')(T'', T')](T', T) = (T''', T).$$

§ 28. Composition of the transformations.

A transformation [§ 27, (9)] is completely determined by the transformation numbers a, b, c, ∂ , and these four integers may be arbitrarily given, provided only that their determinant $n = a\partial - bc$ is positive. If one gives these four numbers the opposite sign, then ω'_1, ω'_2 , by § 27, (6), pass into $-\omega'_1, -\omega'_2$; and if one replaces a, b, c, ∂ by $ma, mb, mc, m\partial$, where m is an arbitrary natural number, then ω'_1, ω'_2 pass into $\omega'_1/m, \omega'_2/m$, and n into m^2n . The period ratio $\omega' = \omega'_2/\omega'_1$ remains unchanged in both cases. For the present, however, we shall always regard two transformations as different when their transformation numbers are different. According to this convention we can denote a transformation unambiguously by a matrix

$$S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}. \quad (1)$$

The determinant

$$n = a\partial - bc \quad (2)$$

is the degree of transformation.

In this notation we also write the relations (6) of § 27 as

$$n(\omega'_1, \omega'_2) = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} (\omega_1, \omega_2). \quad (3)$$

If

$$S' = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix}, \quad a'\partial' - b'c' = n',$$

then

$$n'(\omega''_1, \omega''_2) = \begin{pmatrix} a' & b' \\ c' & \partial' \end{pmatrix} (\omega'_1, \omega'_2), \quad (4)$$

and if in (4) one expresses ω'_1, ω'_2 by (3) in terms of ω_1, ω_2 , one obtains

$$nn'(\omega''_1, \omega''_2) = \begin{pmatrix} a'a + b'c & a'b + b'\partial \\ c'a + \partial'c & c'b + \partial'\partial \end{pmatrix} (\omega_1, \omega_2). \quad (5)$$

If we put, therefore,

$$S'' = S'S, \quad (6)$$

then

$$S'' = \begin{pmatrix} a'a + b'c & a'b + b'\partial \\ c'a + \partial'c & c'b + \partial'\partial \end{pmatrix} = \begin{pmatrix} a'' & b'' \\ c'' & \partial'' \end{pmatrix},$$

and

$$a''\partial'' - b''c'' = n'' = nn'.$$

The transformations S therefore compose according to the same rule as the linear substitutions and matrices which we considered in the sixth section of the second volume. The degree of a compound transformation is equal to the product of the degrees of its components. Here these matrices are tied to the assumption that their elements are integers and their determinant is positive.

These properties are preserved under the composition of transformations. Nevertheless, the totality $\mathfrak{S}$ of transformations S does not form a group, any more than the totality of natural numbers, under composition by multiplication, forms a group; for for given S', S'' one cannot always determine an S satisfying condition (6), as would be required for a group.

By the special transformation of degree m^2 ,

$$M = \begin{pmatrix} \pm m & 0 \\ 0 & \pm m \end{pmatrix},$$

the periods ω_1, ω_2 pass into $\omega'_1 = \omega_1/m$, $\omega'_2 = \omega_2/m$, and the period ratio $\omega = \omega_2/\omega_1$ remains unchanged. These transformations are called multiplications (similarity transformations). Among them is contained the identical substitution

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which leaves everything unchanged and plays the role of the unit in composition.

The multiplications are interchangeable, under composition, with every transformation S :

$$SM = MS. \quad (7)$$

If in SM or MS one holds the transformation S fixed and lets M run through the totality $\mathfrak{M}$ of multiplications, then one obtains a system $\mathfrak{M}S$ which one would, according to Vol. II, § 46, have to call a collineation. If S_1 and S_2 belong to a collineation C , and S'_1 and S'_2 to a collineation C' , then $S'_1 S_1$ and $S'_2 S_2$ also belong to the same collineation C'' . Thus, putting $C'' = C' C$, one can compose the collineations. With this composition the collineation $\mathfrak{M}$ plays the role of the unit. If $S = \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}$ is an arbitrary transformation, then

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}$$

is a multiplication. The two collineations

$$C = \mathfrak{M} \begin{pmatrix} a & b \\ c & \partial \end{pmatrix}, \quad C^{-1} = \mathfrak{M} \begin{pmatrix} \partial & -b \\ -c & a \end{pmatrix}$$

give, therefore, under composition

$$CC^{-1} = C^{-1}C = \mathfrak{M},$$

and are accordingly reciprocal to one another.

Consequently the totality of the collineations forms a group.

The transformations of degree 1 are called linear transformations. For these we denote the transformation numbers, to distinguish them, by the Greek letters $\alpha, \beta, \gamma, \delta$, so that

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1,$$

denotes a linear transformation. The system $\mathfrak{L}$ of linear transformations is a group contained in $\mathfrak{S}$; for if

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad L' = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix},$$

then

$$L'' = L'L = \begin{pmatrix} \alpha'\alpha + \beta'\gamma & \alpha'\beta + \beta'\delta \\ \gamma'\alpha + \delta'\gamma & \gamma'\beta + \delta'\delta \end{pmatrix} = \begin{pmatrix} \alpha'' & \beta'' \\ \gamma'' & \delta'' \end{pmatrix}$$

is also linear, and, for given L', L'' , one can determine L uniquely from the equations

$$\begin{aligned} \alpha'\alpha + \beta'\gamma &= \alpha'', & \alpha'\beta + \beta'\delta &= \beta'', \\ \gamma'\alpha + \delta'\gamma &= \gamma'', & \gamma'\beta + \delta'\delta &= \delta''. \end{aligned}$$

The unit of the group $\mathfrak{L}$ is the identical substitution $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and every substitution L has its reciprocal L^{-1} , as follows from the composition

$$LL^{-1} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The group $\mathfrak{L}$ is infinite and is not commutative.

From the equation

$$\alpha\delta - \beta\gamma = 1 \tag{8}$$

it follows that neither α, β , nor α, γ , nor δ, β , nor δ, γ can have a common factor. Conversely, if α, β have been assumed arbitrarily without a common divisor, then γ, δ can still be determined from (8) in infinitely many ways. If γ, δ is one of these determinations, then all are contained in the form

$$\gamma + \lambda\alpha, \quad \delta + \lambda\beta, \tag{9}$$

where λ is an arbitrary integer.

§ 29. Composition of the transformations from simpler ones.

In the system $\mathfrak{S}$ of all transformations S there is contained a system $\mathfrak{S}_0$, consisting of all transformations S_0 whose second transformation number $b = 0$, while a and ∂ are positive:

$$S_0 = \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}. \tag{1}$$

On composing two S_0 there again arises an S_0 , but nevertheless $\mathfrak{S}_0$ is no more a group than is $\mathfrak{S}$.

One can reduce any arbitrary transformation

$$S = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

by a composition LS to an S_0 . For if

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \tag{2}$$

is to hold, then α, β must satisfy

$$\alpha q + \beta s = 0,$$

and hence, if ∂ is the greatest common divisor of q and s , one puts

$$\partial\alpha = s, \quad \partial\beta = -q,$$

and determines, after α and β have thereby been found as relative primes, γ and δ from formula (8) of § 28. Then (2) is fulfilled if

$$a\partial = n, \quad c = \gamma p + \delta r,$$

is put; a and ∂ are thereby determined uniquely, but with another choice of γ and δ , c can be replaced by $c + \lambda a$. One can therefore choose λ so that c is contained in the series of numbers

$$0, 1, 2, \dots, a - 1,$$

and thereby the substitution S_0 is completely determined.

If the four transformation numbers have a common factor, then this can be separated off by means of the formula

$$\begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \begin{pmatrix} a & b \\ c & \partial \end{pmatrix} = \begin{pmatrix} ma & mb \\ mc & m\partial \end{pmatrix}$$

by composition with the multiplication; hence we now assume that a, b, c, ∂ have no common divisor. One can then always determine two integers ξ, η such that

$$a\eta - c\xi = \alpha, \quad b\eta - \partial\xi = \beta \quad (3)$$

have no common divisor.

To see this, we first assume ξ, η relatively prime. Then every common divisor of α, β is necessarily a divisor of n , as is seen from the solutions of (3),

$$n\xi = b\alpha - a\beta, \quad n\eta = \partial\alpha - c\beta. \quad (4)$$

Thus one takes ξ not divisible by all the prime numbers which occur simultaneously in a and b , but ξ divisible, η indivisible, by all the other prime numbers occurring in n , and moreover ξ, η relatively prime, which is always possible. Then α and β have no common divisor. Thereupon one determines γ, δ so that

$$\alpha\delta - \beta\gamma = 1.$$

Then, by (3), also

$$(a\delta - b\gamma)\eta - (c\delta - \partial\gamma)\xi = 1,$$

and the following composition results, as is easily recognized with the aid of (4):

$$\begin{pmatrix} a & b \\ c & \partial \end{pmatrix} = \begin{pmatrix} a\delta - b\gamma & \xi \\ c\delta - \partial\gamma & \eta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \quad (5)$$

If we call

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \quad (6)$$

the principal transformation of degree n , then it is proved that all transformations of degree n can be composed from a multiplication, a principal transformation, and linear transformations.

From the composition

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & m \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & mn \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}, \quad (7)$$

we can still further conclude that every transformation of degree n can be composed from such transformations whose degree is a prime number. If $n = pq$ is decomposed into two factors p and q , which are relatively prime to one another, then, by determining the numbers β, δ from

$$p\delta - q\beta = 1,$$

the composition

$$\begin{pmatrix} p & \beta \\ q & \delta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix} \begin{pmatrix} p\delta & -q\beta \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$$

is obtained, from which it is seen that, instead of the principal transformation, one may also use any of these transformations $\begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix}$ for the derivation of all the others.

§ 30. The linear fundamental transformations.

The whole group $\mathfrak{L}$ of linear transformations can be derived by repetition of two among them, which we call the linear fundamental transformations.

If

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1,$$

is an arbitrary linear transformation, then

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (1)$$

and, since the identical transformation is the unit in the group L ,

$$L^{-1} = \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

is the transformation reciprocal to L .

We denote by the power $L^{\pm m}$ what arises by m -fold repetition of L or L^{-1} , and will now prove that, by the powers of the fundamental transformations

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (2)$$

every substitution L of the group $\mathfrak{L}$ can be composed. To begin with,

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad A^\lambda = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \quad (3)$$

(for every integral positive or negative λ), and

$$B^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B^3 = B^{-1}. \quad (4)$$

We also put

$$C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = B^{-1}AB, \quad C^\lambda = \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix}, \quad (5)$$

so that C is derivable from A and B .

Now let

$$L = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

be an arbitrary linear transformation. We derive from it the series

$$L' = LA^\lambda, \quad L'' = L'C^{\lambda'}, \quad L''' = L''A^{\lambda''}, \quad L'''' = L'''C^{\lambda'''},$$

whose first and second elements are formed as follows:

$$\alpha' = \alpha + \lambda\beta, \quad \beta'' = \beta' - \lambda'\alpha', \quad \alpha''' = \alpha'' + \lambda''\beta'', \quad \beta'''' = \beta''' - \lambda'''\alpha''', \dots,$$

and one can dispose of $\lambda, \lambda', \lambda'', \lambda''', \dots$ so that, in absolute value,

$$\alpha' \leq \frac{1}{2}\beta, \quad \beta'' \leq \frac{1}{2}\alpha', \quad \alpha''' \leq \frac{1}{2}\beta'', \quad \beta'''' \leq \frac{1}{2}\alpha''', \dots,$$

as long as none of these numbers vanishes. The numbers

$$\beta, \alpha', \beta'', \alpha''', \beta''', \dots$$

therefore form a sequence decreasing in absolute value, and after a finite number of compositions of this kind one number of this series must vanish.

If $\beta^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} \pm 1 & 0 \\ \gamma^{(\nu)} & \pm 1 \end{pmatrix} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} A^{\pm \gamma^{(\nu)}},$$

and if $\alpha^{(\nu)} = 0$, then

$$L^{(\nu)} = \begin{pmatrix} 0 & \pm 1 \\ \mp 1 & \delta^{(\nu)} \end{pmatrix} = A^{\pm \delta^{(\nu)}} B^{\pm 1}.$$

And since

$$L = L' A^{-\lambda} = L'' C^{-\lambda'} A^{-\lambda} = L''' A^{-\lambda''} C^{-\lambda'} A^{-\lambda} = \dots,$$

the theorem is proved.

§ 31. The linear fundamental transformations of the ϑ -functions.

In the application to the transformation of the Θ -functions [§ 27, (12)], the transformed modulus

$$\omega' = \frac{c + \partial \omega}{a + b \omega}$$

first comes under consideration. It does not change when the four transformation numbers receive a common factor m . The transformation is called a proper one if a, b, c, ∂ have no common factor.

The transformed argument

$$u' = \frac{nu}{a + b \omega}$$

passes into mu' , when a, b, c, ∂ are replaced by $ma, mb, mc, m\partial$, and hence n by $m^2 n$. The multiplication $\begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}$ leaves the modulus ω unchanged and changes u into mu .

According to the results of the preceding two paragraphs, the whole system of proper transformations can be derived by repeated application of the three transformations

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix},$$

and we therefore first consider the linear fundamental transformations of the ϑ -functions.

I. $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ or $(\omega, \omega + 1)$.

According to § 27, (11), (12), one has

$$\vartheta_{11}(u, \omega + 1) = A \vartheta_{11}(u, \omega), \tag{1}$$

where A is independent of u . If one replaces u by

$$u + \frac{1}{2}, \quad u + \frac{\omega}{2}, \quad u + \frac{1 + \omega}{2},$$

then the formulae § 21, (8), give

$$\vartheta_{10}(u, \omega + 1) = A \vartheta_{10}(u, \omega), \tag{2}$$

$$e^{\pi i/4} \vartheta_{00}(u, \omega + 1) = A \vartheta_{01}(u, \omega), \tag{3}$$

$$e^{\pi i/4} \vartheta_{01}(u, \omega + 1) = A \vartheta_{00}(u, \omega). \tag{4}$$

To determine the constant A we use, as will often happen in what follows, the method of putting $u = 0$ in (1) after differentiation, and then making use on the right and left of formula (5), § 23,

$$\vartheta'_{11} = \pi \vartheta_{00} \vartheta_{10} \vartheta_{01}; \quad (5)$$

one obtains

$$A^2 = e^{\pi i/2}, \quad A = e^{\pi i/4}.$$

That the positive sign stands with A follows from one of formulae (3), (4), according to the concluding remark of § 25, when one lets ω become infinite. Consequently one obtains

$$\begin{aligned} \vartheta_{11}(u, \omega + 1) &= e^{\pi i/4} \vartheta_{11}(u), \\ \vartheta_{10}(u, \omega + 1) &= e^{\pi i/4} \vartheta_{10}(u), \\ \vartheta_{01}(u, \omega + 1) &= \vartheta_{00}(u), \\ \vartheta_{00}(u, \omega + 1) &= \vartheta_{01}(u). \end{aligned} \quad (6)$$

$$\text{II. } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \text{ or } \left(\omega, -\frac{1}{\omega} \right).$$

Again by § 27, (12),

$$e^{-\pi i u^2 / \omega} \vartheta_{11} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) = A \vartheta_{11}(u, \omega), \quad (7)$$

and by increasing u by $1/2$, $\omega/2$, $(1 + \omega)/2$,

$$e^{-\pi i u^2 / \omega} \vartheta_{01} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) = i A \vartheta_{10}(u, \omega), \quad (8)$$

$$e^{-\pi i u^2 / \omega} \vartheta_{10} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) = i A \vartheta_{01}(u, \omega), \quad (9)$$

$$e^{-\pi i u^2 / \omega} \vartheta_{00} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) = i A \vartheta_{00}(u, \omega). \quad (10)$$

From this, as above, one obtains

$$A = \pm i \sqrt{-i\omega}.$$

From $u = 0$ and a purely imaginary ω one concludes that, if $\sqrt{-i\omega}$ is taken so that its real part is positive, the lower sign must stand, and hence

$$\begin{aligned} e^{-\pi i u^2 / \omega} \vartheta_{11} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) &= -i \sqrt{-i\omega} \vartheta_{11}(u), \\ e^{-\pi i u^2 / \omega} \vartheta_{01} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) &= \sqrt{-i\omega} \vartheta_{10}(u), \\ e^{-\pi i u^2 / \omega} \vartheta_{10} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) &= \sqrt{-i\omega} \vartheta_{01}(u), \\ e^{-\pi i u^2 / \omega} \vartheta_{00} \left(\frac{u}{\omega}, -\frac{1}{\omega} \right) &= \sqrt{-i\omega} \vartheta_{00}(u). \end{aligned} \quad (11)$$